Lesson notes · for you

Loops: for and while

Programming · Level 2 · Lesson 4

Time
20 min

Materials
A phone with this lesson downloaded; Squared paper for patterns

Learning goals

- Repeat with for and range.
- Count and add up in a loop.
- Use while, and avoid a loop that never ends.

Activity: Be the loop

1. Warm-up (3 min): How long would it take to write print(1) to print(100)? Now write the loop that does it in two lines.
2. Act (7 min): In groups of three: one is the loop and reads range(1, 6) aloud, one is i and holds up each number, one is total and adds it on paper.
3. Predict (7 min): Change it to range(1, 11) and range(5). Predict the total each time before you act it out.
4. Run (3 min): Run the programs on the phone. Were your totals right?

Acting it out works without a phone; one phone is enough to check at the end.

Discussion

- Why does range stop before its second number?
- When would you use while instead of for?
- How could a loop help you search a list?

Check yourself

- Write a loop that prints 10, 20, 30, 40, 50.

Try it this week
Write a loop that prints a triangle of stars: for i in range(1, 6): print("*" * i). Then make it upside down.

Worksheet

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Name _____ Class _____ Date _____

1 **What is total at the end?**

```
total = 0
for n in range(1, 4):
    total = total + n
print(total)
```

◯ 6 ◯ 10 ◯ 4

2 **Finish the program: it asks which times table and how many lines, then prints them, like 9 x 1 = 9.**

```
t = int(input("Table? "))
k = int(input("How many lines? "))
# print t x 1 = ... up to t x k
```

Write your program:

3 **Fill the gaps so the program prints the numbers 1 to 10.**

```
for n in range( _____ , _____ ):
    print(n)
```

Word bank: `0` `10` `1` `11`

4 **What does this program print?**

```
n = 1
while n < 20:
    n = n * 2
print(n)
```

◯ 16 ◯ 20 ◯ 32

5 **This loop never ends.**

```
n = 1
while n > 0:
    n = n + 1
```

◯ True ◯ False

✎ **Which boring job could a loop do for you: writing lines, counting, adding? Write the loop in words.**

Check your answers

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1 **6**

range(1, 4) gives 1, 2 and 3, so total is 1 + 2 + 3 = 6.

2

```
t = int(input("Table? "))
k = int(input("How many lines? "))
for i in range(1, k + 1):
    print(t, "x", i, "=", t * i)
```

range(1, k + 1) gives 1 to k, because range stops before its second number.

3

```
for n in range(1, 11):
    print(n)
```

range starts at its first number and stops before its second, so 1 to 10 is range(1, 11).

4 **32**

n goes 1, 2, 4, 8, 16, 32. At 16 the condition is still True, so it doubles once more to 32; then 32 < 20 is False and the loop stops.

5 **True**

n starts at 1 and only gets bigger, so n > 0 is always True.